

Katherine Jackson

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EDUCATION

Carnegie Mellon University (Pittsburgh, PA)

Spring 2026

Bachelor of Science, [Electrical and Computer Engineering](#)

Minor, Biomedical Engineering

- **Programming Languages:** C/C++, JavaScript, MATLAB, Python, SQL, SystemVerilog
- **Hardware Skills:** ADC Sampling, AutoDesk Fusion, Finite State Machines, I2C, PID Controls, UART
- **Frameworks and Libraries:** Git, Linux, NumPy, MATLAB Simulink, PyTorch, Scikit-learn
- **Other Relevant Coursework:** Computer Systems, Computer Vision

WORK EXPERIENCE

Electrical Engineer Intern | [Quintech Electronics & Communication](#) (Pittsburgh, PA)

Summer 2025

- Performed network analysis to increase strength of signal and performance of radio frequency (RF) PCBs by adjusting capacitance, resistance and channel paths
- Decreased re-manufacturing time by identifying and correcting short-circuit patterns on prototype high-frequency boards

PROJECT EXPERIENCE

[American Sign Language \(ASL\) Reading Glove Sensor](#) | Senior Design Project

- Engineered a Python handler script to collect and wirelessly transmit flex, touch sensor, and IMU data from a Raspberry Pi Pico via I2C connection and ADC sampling
- Implemented K-Nearest Neighbor, Random Forest and Neural Network classifiers utilizing a feature vector to interpret complex hand movements into digital text with 92% validation accuracy

Neural Networks & Natural Language Processing | Machine Learning

- From scratch, implemented and trained a Neural Network classifier using stochastic gradient descent and backpropagation with manually derived gradients
- Developed an autonomous decision-making model using Markov Decision Processes (MDPs) and Q-learning to optimize rewards for simulated environments

Motor Control using RTOS | Embedded Systems

- Developed a multi-threaded application on an STM32 (ARM Cortex-M4) using RTOS to manage PID motor control, UART communication, and real-time LCD visualization
- Optimized car motor response by using hardware interrupts for PID updates, reducing speed error

Bluetooth Encoded Receiver | Digital Systems

- Designed a SystemVerilog SECCED decoder to provide single-error correction and two-bit error detection for reliable data transmission
- Engineered a Finite State Machine (FSM) to interface with asynchronous Bluetooth UART signals, enabling wireless communication at variable baud rates

LEADERSHIP

NCAA Division III Student-Athlete & Team Captain | [Carnegie Mellon Volleyball](#)

2022 - 2025

- Captain of Varsity Volleyball Team; in 2025, led team to their first ever conference championship title and second NCAA Sweet Sixteen appearance in program history
- Dedicated 20 hours a week to training and competing while carrying a full academic course load as a four year student-athlete

Instructor | [Society of Women Engineers](#), Carnegie Mellon University

Spring 2024

- Taught Python fundamentals to ~20 middle school girls each weekend in the semester
- Produced a lesson plan ahead of each session and provided one-on-one guidance